

Deep Learning: Theory and Practice

Linear and Logistic Models for Classification

01-02-2018

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Matrix Derivatives

- ❖ Derivative of a scalar with a vector

$$\left(\frac{\partial x}{\partial \mathbf{a}}\right)_i = \frac{\partial x}{\partial a_i}$$

- ❖ Derivative of a vector with a scalar

$$\left(\frac{\partial \mathbf{a}}{\partial x}\right)_i = \frac{\partial a_i}{\partial x}$$

- ❖ Derivative of a vector with a vector

$$\left(\frac{\partial \mathbf{a}}{\partial \mathbf{b}}\right)_{ij} = \frac{\partial a_i}{\partial b_j}.$$

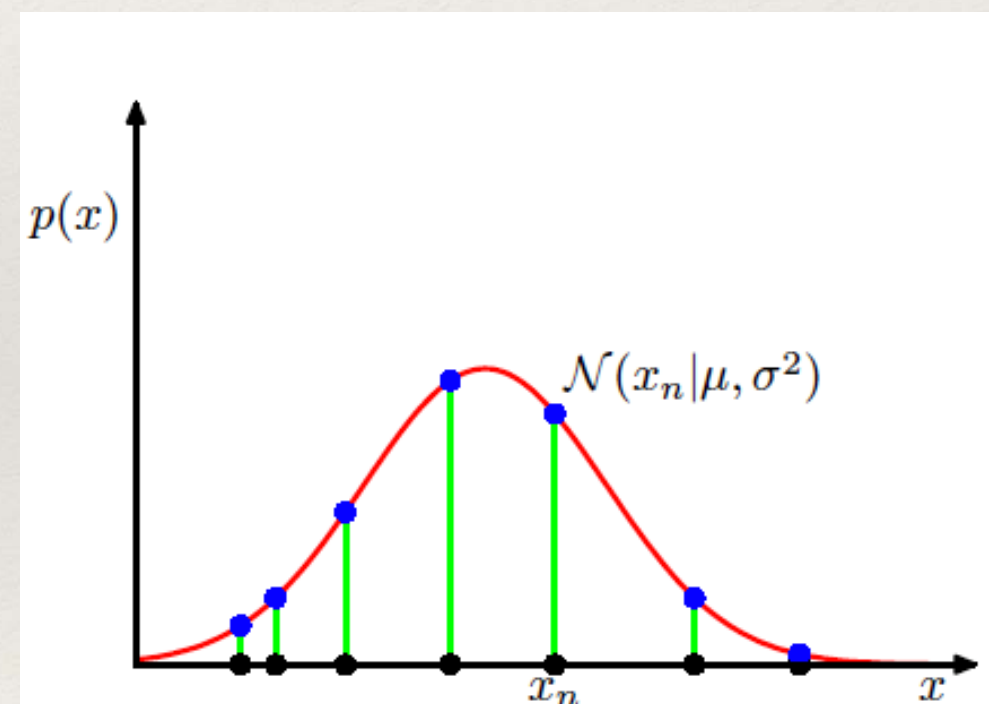
- ❖ Derivative of trace of a matrix ?

Maximum Likelihood

❖ Gaussian Distribution - multivariate

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$$p(\mathbf{x}|\mu, \sigma^2) = \prod_{n=1}^N \mathcal{N}(x_n|\mu, \sigma^2)$$



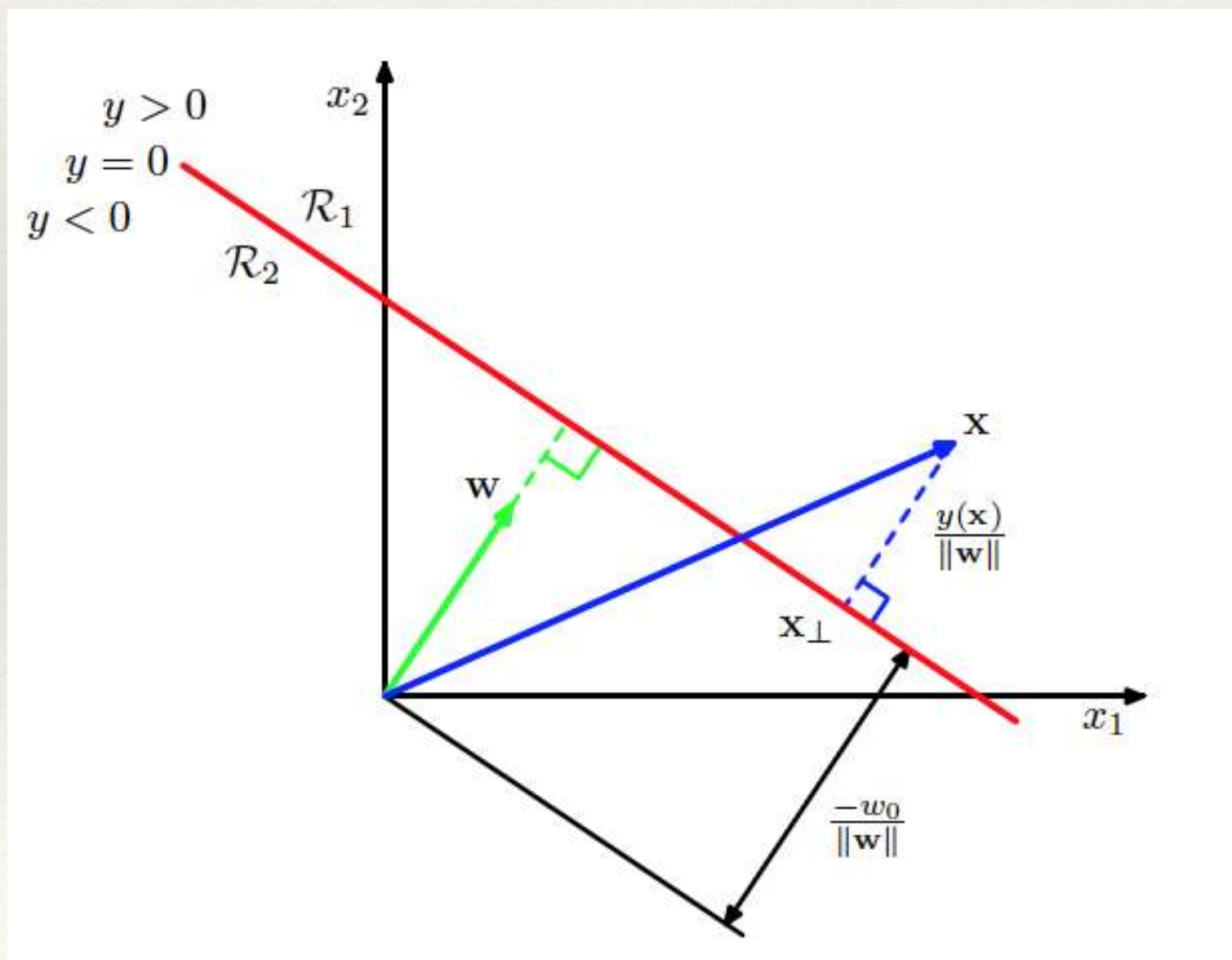
$$\ln p(\mathbf{x}|\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2} \ln \sigma^2 - \frac{N}{2} \ln(2\pi)$$

$$\mu_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N x_n$$

Linear Models for Classification

- ❖ Optimize a modified cost function

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$$



Least Squares for Classification

- ❖ K-class classification problem

$$y_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

$$\mathbf{y}(\mathbf{x}) = \widetilde{\mathbf{W}}^T \widetilde{\mathbf{x}}$$

- ❖ With 1-of-K hot encoding, and least squares regression

$$E_D(\widetilde{\mathbf{W}}) = \frac{1}{2} \text{Tr} \left\{ (\widetilde{\mathbf{X}} \widetilde{\mathbf{W}} - \mathbf{T})^T (\widetilde{\mathbf{X}} \widetilde{\mathbf{W}} - \mathbf{T}) \right\}$$

Logistic Regression

- ❖ 2- class logistic regression

$$p(C_1|\phi) = y(\phi) = \sigma(\mathbf{w}^T \phi)$$

- ❖ Maximum likelihood solution

$$\nabla E(\mathbf{w}) = \sum_{n=1}^N (y_n - t_n) \phi_n$$

- ❖ K-class logistic regression

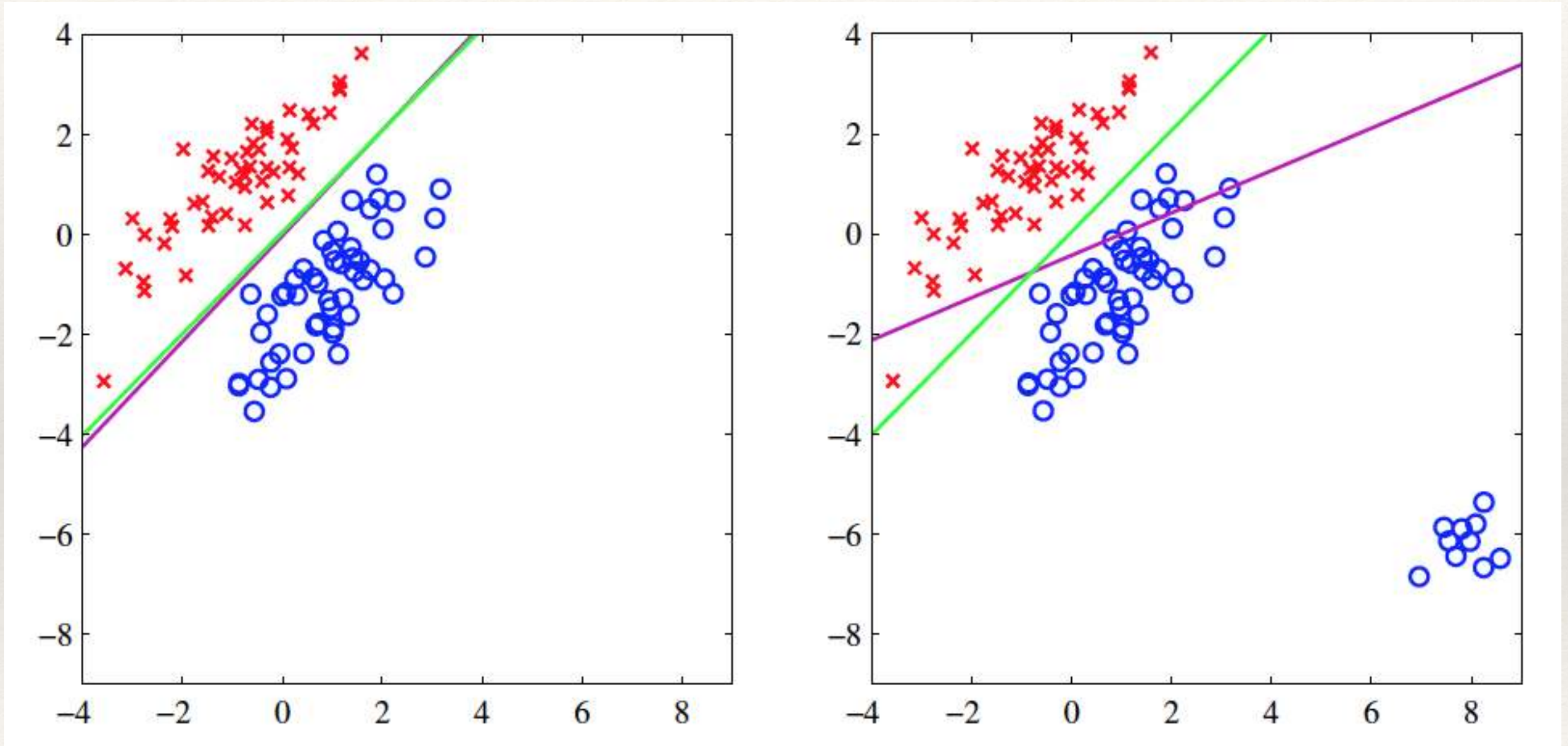
$$p(C_k|\phi) = y_k(\phi) = \frac{\exp(a_k)}{\sum_j \exp(a_j)}$$

- ❖ Maximum likelihood solution

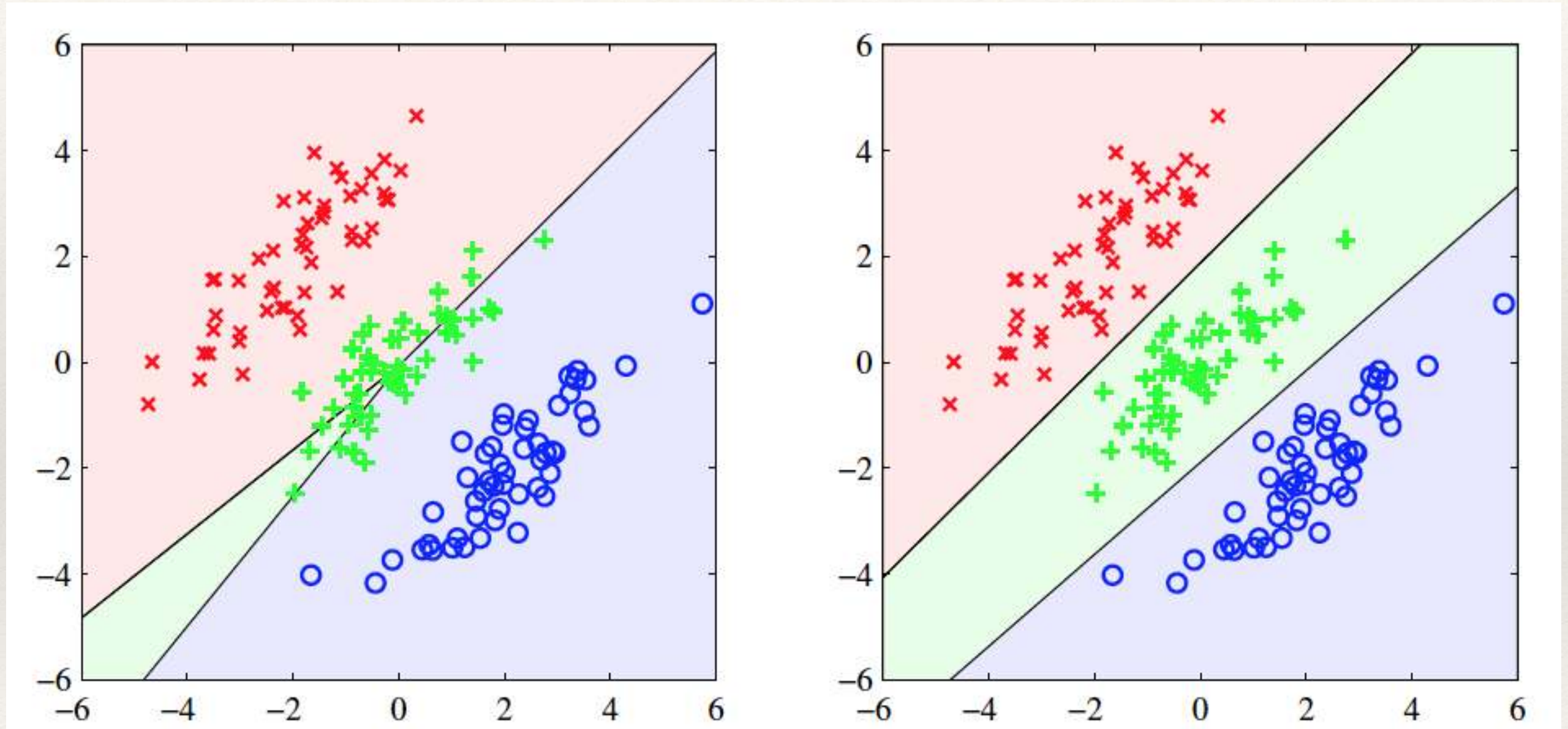
$$a_k = \mathbf{w}_k^T \phi.$$

$$\nabla_{\mathbf{w}_j} E(\mathbf{w}_1, \dots, \mathbf{w}_K) = \sum_{n=1}^N (y_{nj} - t_{nj}) \phi_n$$

Least Squares versus Logistic Regression

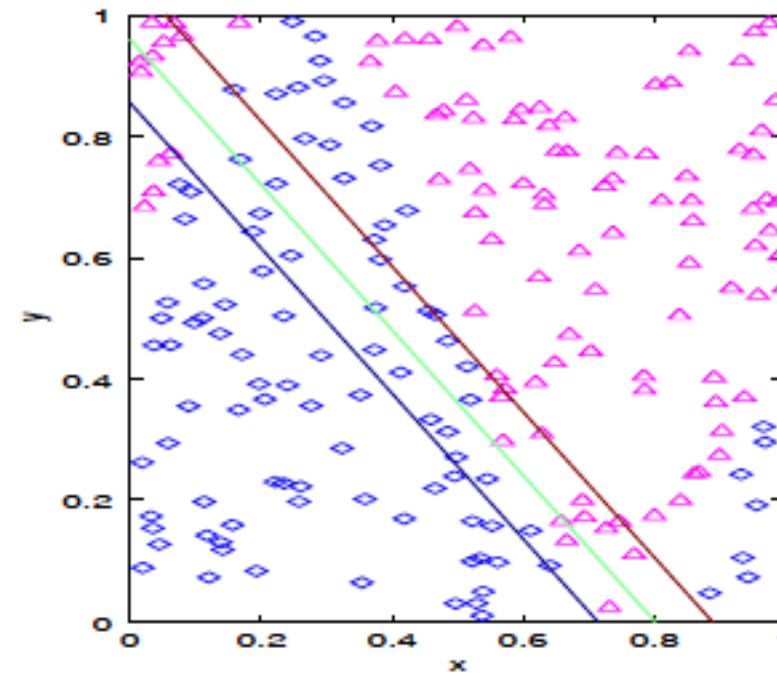
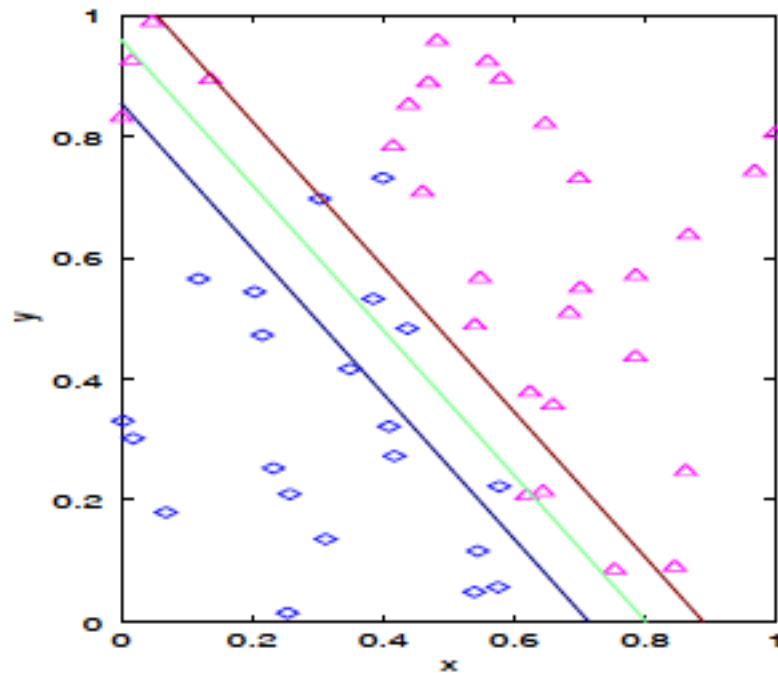


Least Squares versus Logistic Regression



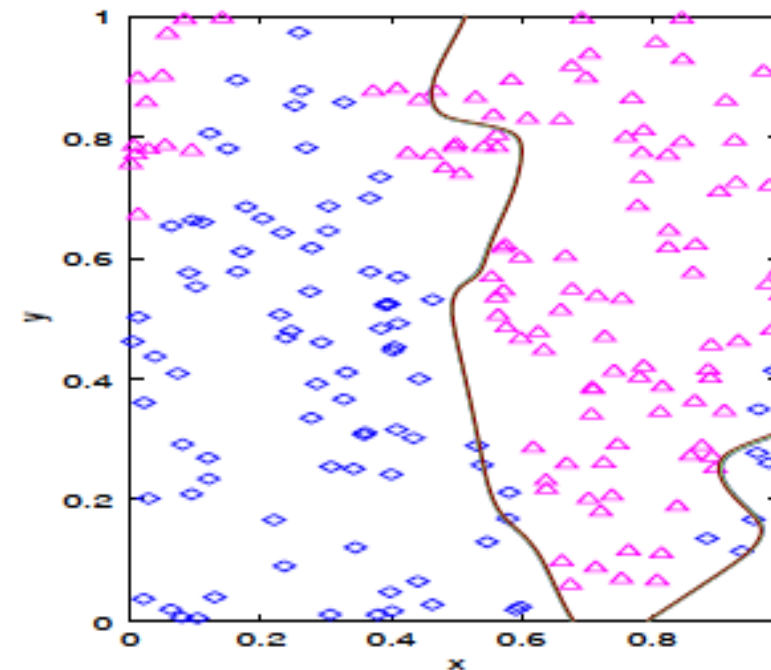
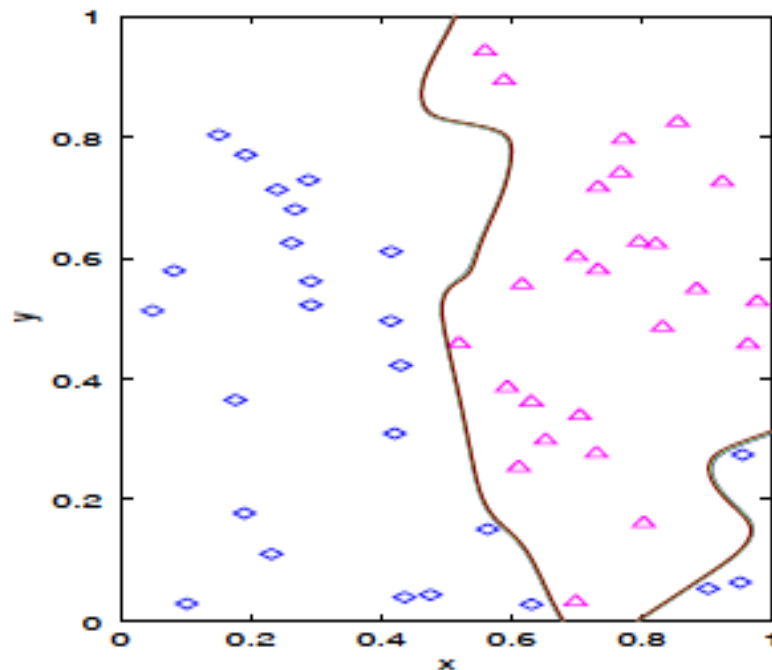
Bishop - PRML book (Chap 4)

Underfit



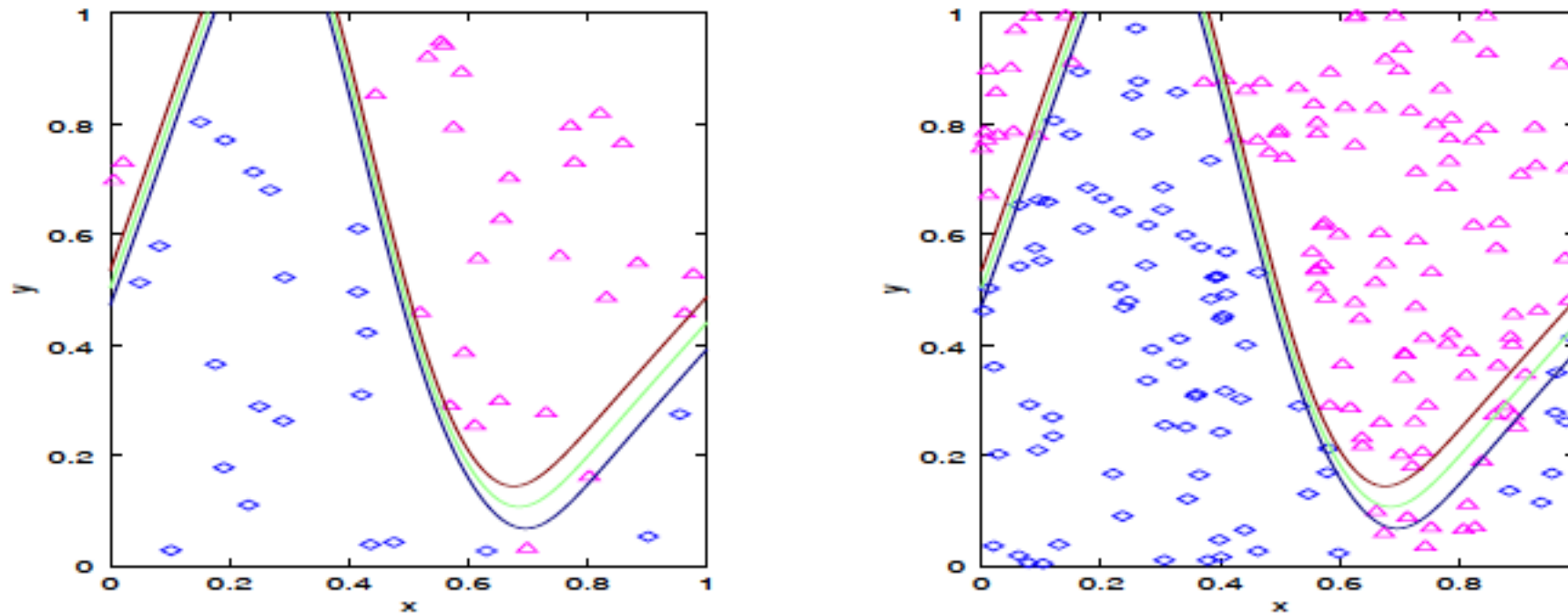
- The model is not able to capture the variability in the data (Linear Model)
- Both the training and testing error are high (15%,20%)
- Try to learn a more complex model – more features, more hidden neurons, decrease regularization
- More data would not help

Overfit



- The model is capturing data as well as accidental variations (100 hidden neurons)
- Training error is too low and testing error is too high (0%, and 16%)
- Try to learn a simpler model – less features, less hidden neurons, increase regularization
- More data would help

Compromise



- Reasonable training and test errors – (4%, 8%)
- Appropriate model – capturing only the global characteristics not details